Blockchain Data Availability: Benchmarking Throughput, Economics, Utilization, and Security of Six Production Protocols

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Abstract

Data availability (DA) is a critical bottleneck in blockchain scalability. This paper presents a benchmarking study of six production DA protocols (Polkadot ELVES, Ethereum EIP-4844/PeerDAS, Celestia, Espresso Tiramisu, NEAR, and Avail) using 90 days of mainnet data (November 2025 to February 2026). We introduce a taxonomy that classifies DA architectures into three paradigms: data availability sampling (DAS), verifiable information dispersal (VID), and validator-set recovery. Under a unified methodology we measure throughput (MiB/s), cost ($/MiB), utilization, and block dynamics across all six protocols. Key findings are: (i) mean utilization stays below 40% for every protocol, so current measurements are demand-constrained rather than supply-constrained; (ii) median DA cost spans more than eight orders of magnitude, from nano-dollars (Espresso) to tens of cents per MiB (NEAR); (iii) no protocol slashes for pure data withholding, so enforcement is indirect in every design; and (iv) no protocol guarantees permanent availability: protocol-defined retention ranges from ∼25 hours (Polkadot) to ∼18 days (Ethereum), while Avail and Espresso leave retention to node operators. Ethereum's Fusaka upgrade and two blob-parameter-only forks fall inside the window and serve as natural experiments on fee-market behaviour. We release all collection tooling, datasets, and analysis code under open licenses.

Keywords: data availability, blockchain scalability, erasure coding, data availability sampling, benchmarking, rollups, distributed systems, measurement

1. Introduction

Blockchain scalability hinges on a deceptively simple requirement: every participant must be confident that block data has been published and is retrievable. This property, known as *data availability* (DA), underpins the security of rollups, parachains, and sharded architectures alike. Without DA guarantees, validators cannot verify state transitions, light clients cannot detect fraud, and the trust model of the entire system collapses to that of a permissioned committee [1].

The past three years have witnessed a proliferation of DA solutions. Ethereum introduced blob-carrying transactions (EIP-4844) and, with the Fusaka upgrade in December 2025, activated PeerDAS for cell-level sampling [2, 3]. Celestia launched the first dedicated DA network with two-dimensional erasure coding and namespace-filtered sampling [4]. Polkadot's ELVES protocol distributes erasure-coded parachain data across its validator set with approval-gated finality [5]. Espresso's Tiramisu introduces a three-layer Verifiable Information Dispersal (VID) architecture that explicitly rejects DAS in favor of algebraic dispersal proofs [6]. NEAR provides sharding-based DA through Nightshade with a rollup-oriented blob store [7], and Avail combines KZG polynomial commitments with DAS-enabled light clients on a Substrate-based chain [8].

Despite this activity, the field lacks an empirical, cross-protocol comparison grounded in production data. Existing comparisons are qualitative surveys [1, 9], vendor-authored, or measurements of a single protocol [10, 11]. No study has simultaneously measured throughput, cost, and utilization across multiple DA networks under a unified methodology. This gap is consequential: rollup developers choosing a DA backend, protocol designers evaluating trade-offs, and researchers assessing the state of the art all lack a common empirical baseline.

This paper fills that gap. Our contributions are:

1. **A taxonomy of DA architectures** that classifies protocols along three paradigms (Data Availability Sampling, Verifiable Information Dispersal, and validator-set recovery) and along orthogonal dimensions of encoding scheme and commitment type (Section 2).
2. **A qualitative architecture comparison** of encoding redundancy, commitment schemes (KZG vs. NMT vs. Merkle), and verification mechanisms across all six protocols, grounded in protocol specifications and documentation (Section 3).
3. **An empirical benchmarking study** using 90 days of production mainnet data (November 2025–February 2026) across Polkadot, Ethereum, Celestia, Espresso, NEAR, and Avail, measuring throughput (MiB/s), cost ($/MiB), utilization (%), and block dynamics under consistent definitions (Section 5).
4. **A comparative security analysis** of trust assumptions,

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Byzantine fault tolerance thresholds, data withholding enforcement, and retention policies across all six protocols (Section 6).

5. **Open-source tooling and datasets**: all data collection scripts, raw datasets, and analysis notebooks are released under Apache 2.0 (code) and CC BY 4.0 (data) licenses to enable reproducibility and longitudinal extension.

2. Background and Related Work

2.1. Data Availability Fundamentals

Data availability (DA) must be distinguished from the related but distinct problems of data *storage* and data *retrieval*. Storage concerns the persistence of data over time; retrieval concerns the ability to read stored data on demand. DA addresses a different question: *was the data published at all?* Formally, a block producer who withholds part of a block's data can cause validators and light clients to accept an invalid state transition, because the withheld data may contain a fraudulent transaction whose invalidity can never be proven if the data is unavailable [12].

Definition 1 (Data Availability)**.** *A block B satisfies the data availability property if every honest participant can retrieve the complete block data within a bounded time window Δ after the block header is published.*

A DA scheme must provide four properties:

1. **Completeness.** All data constituting block B is published to the network. A DA scheme is complete if no honest node accepts a block header unless the corresponding data has been made available to the network.
2. **Soundness.** The encoding of B is verifiably correct. Given a commitment $C(B)$ included in the block header, any participant can verify that a received coded piece is consistent with $C(B)$ without possessing the entire block.
3. **Availability detection.** Resource-constrained participants (light clients) can assess whether the data behind a block header is available without downloading the full block. This property separates DA from simple data retrieval.
4. **Reconstruction.** The original data B can be recovered from any sufficiently large subset of coded pieces. If $E(B)$ denotes the erasure-coded encoding of B producing n coded pieces with redundancy factor r, then B is reconstructible from any n/r pieces.

Notation. Throughout this paper we use the following conventions. A block's raw data is denoted B, with $|B|$ its size in bytes. The encoding function $E(B)$ produces n coded pieces (variously called shares, chunks, or cells depending on the protocol). A commitment $C(B)$ is included in the block header and serves as a binding digest of $E(B)$. Verification $V(C, s_i)$ checks that coded piece s_i is consistent with commitment C. The redundancy factor $r = n \cdot |s_i|/|B|$ captures the total storage expansion introduced by the encoding.

2.2. Taxonomy of DA Approaches

We classify the six protocols along three dimensions: the *availability verification paradigm* (how clients gain confidence that data is available), the *encoding scheme* (how raw data is expanded into coded pieces: 1D Reed–Solomon [13], 2D Reed–Solomon, or polynomial VID), and the *commitment scheme* (how pieces are bound to the header: KZG [14], Namespaced Merkle Trees, or plain Merkle trees). Section 3 compares encodings and commitments in detail; here we define the three verification paradigms.

DAS (Avail, Celestia, Ethereum/PeerDAS). Light clients request a few random coded pieces; if all arrive with valid proofs, the client concludes with high probability that the full data is available. Security rests on an *honest-minority* assumption: at least one honest, non-eclipsed full node must seed correctly encoded pieces into the network.

Validator-set recovery (Polkadot, NEAR). The validator set collectively stores erasure-coded chunks and attests to availability through consensus (Polkadot's availability bitfields [5], NEAR's per-shard chunk producers [7]); there is no client-side sampling.

VID (Espresso). Verifiable information dispersal [15] combines erasure coding with algebraic proofs that each committee member received a valid piece; a threshold of signatures forms a DA certificate [6]. Light-client confidence derives from the certificate, not from sampling.

2.3. Protocols Under Study

We summarize each protocol as deployed during our window (November 2025–February 2026); Table 3 gives the parameters.

Polkadot (ELVES). The relay chain erasure-codes parachain Proof-of-Validity (PoV) data across ~600 validators [5, 16]; a block is available once two-thirds sign availability bitfields, after which it enters approval checking and GRANDPA finality. Up to 100 cores each carry a PoV of up to 10 MiB every 6 s. DA is priced through coretime rather than per byte, and chunks are kept for ~25 hours.

Ethereum (EIP-4844 and PeerDAS). Blobs of 128 KiB are committed with KZG and priced by a separate exponential blob-gas market [2]. Pectra (EIP-7691 [17]) set a target/maximum of 6/9 blobs per block. The Fusaka upgrade on 3 December 2025 activated PeerDAS (EIP-7594 [3]), with 2× column extension over 128 columns, and EIP-7918 [18], which bounds the blob base fee from below by a fraction of the execution base fee. Two blob-parameter-only (BPO) forks then raised the target/maximum to 10/15 (9 December) and 14/21 (7 January 2026). Blobs are pruned after 4,096 epochs (~18 days).

Celestia. A $k \times k$ original data square is extended to $2k \times 2k$ and committed with Namespaced Merkle Trees, enabling namespace-filtered retrieval [4]. CometBFT gives single-slot finality every ∼6 s. App version 6 activated on 24 November 2025, and on 3 December the maximum square rose from 128×128 (8 MiB) to 256×256 (32 MiB). Light nodes sample within a 7-day window (CIP-036).

Espresso (Tiramisu). Tiramisu layers a KZG-based VID scheme (Savoiardi), a DA committee for fast retrieval (Mascarpone), and an untrusted CDN for bulk delivery [6]; it is now marketed as EspressoDA. HotShot gives responsive finality of ∼2 s and the maximum block size was 1 MB throughout our window. Mainnet 0 runs 100 nodes operated by 20 parties without slashing. Retention is set per node: pruning is off by default and, when enabled, targets 7 days.

NEAR. Nightshade shards erasure-code chunk data among each shard's chunk producers [7]. Rollups post data as function-call input to a blob-store contract; it is pruned after the 5-epoch garbage-collection window (∼1.5 days at the ∼0.6 s block time in force since May 2025). A single transaction is capped at 1.5 MiB. There is no DAS and no active withholding challenge.

Avail. Row-wise 1D Reed–Solomon encoding with KZG commitments supports DAS by light clients over a DHT [8]. BABE/GRANDPA produce a block every 20 s; application-specific data retrieval lets clients download only their application's data. The maximum block size was 4 MiB throughout the window. Nodes prune *state* after 256 blocks but keep block data by default, so there is no protocol-defined DA expiry; the VectorX bridge relays DA attestations to Ethereum with a ∼2-hour delay.

2.4. Related Work

Foundational theory. Croman et al. [19] quantified the throughput and latency limits of monolithic blockchains, motivating the separation of data publication from execution. Al-Bassam et al. [12] introduced fraud and data availability proofs, the foundation for DAS: by erasure-coding block data and letting light clients randomly sample coded pieces, a network can guarantee availability with high probability under an honest-minority assumption. Hall-Andersen, Simkin, and Wagner formalized the security definition of DAS [20] and proposed FRIDA, a DAS construction without a trusted setup [21]. Coded Merkle Trees [22] and the ACeD oracle [23] reduce the cost of detecting incorrect coding. On the dispersal side, asynchronous verifiable information dispersal [15] is the primitive behind committee-based designs, and Nazirkhanova, Neu, and Tse [24] adapted it to rollups using KZG commitments [14]; Fisch et al. [25] extend VID to permissionless settings.

Protocol-specific designs. Al-Bassam's LazyLedger proposal [4] introduced a dedicated DA blockchain with namespace-filtered retrieval, which evolved into Celestia. Burdges et al. [5] formalized Polkadot's ELVES protocol for availability and validity, proving security under Byzantine assumptions with approval-based checking. Bearer et al. [6] describe the HotShot consensus and Tiramisu DA architecture. Ethereum has progressed from blob transactions (EIP-4844 [2]) through blob throughput increases (EIP-7691 [17]) to peer-level DAS (EIP-7594 [3]), an incremental path toward Danksharding [26]; Chaudhuri et al. [27] and Pigaglio et al. [28] study the networking design space of Ethereum DAS through simulation.

Table 1: Encoding and redundancy.

Protocol	Encoding	Redund.	Reconstruction
Polkadot [5]	1D RS (validators)	∼3×	f+1 of n (n=3f+k)
Ethereum [3]	1D RS (PeerDAS cells)	∼2×	50%+ of 128 columns
Celestia [4]	2D RS ($k \times k \to 2k \times 2k$)	∼4×	k×k shares
Espresso [6]	Polynomial VID	$1/r$, r=m/n	m of n evaluations
NEAR [7]	1D RS (chunk parts)	∼3×	1/3 of parts
Avail [8]	1D RS (row-wise)	∼2×	n of $2n$ coded pieces

Empirical and economic studies. Several recent studies measure Ethereum's blob market in isolation. Crapis et al. [29] model the economics of EIP-4844; Heimbach et al. [10] and Park et al. [30] analyse blob fee dynamics and the impact of EIP-4844; and Lee et al. [11] measure blob utilisation over 180 days. All of these focus on Ethereum alone.

Surveys. Saif et al. [1] and Thibault et al. [9] survey DA and rollup designs, classifying approaches by trust assumptions and encoding strategies, without measurements from production networks.

Positioning of this work. To our knowledge, this paper is the first to measure six production DA protocols simultaneously under one methodology. Prior empirical work covers Ethereum alone, and prior comparative work is qualitative or simulation-based. Our study complements the theoretical results above and the surveys of [1, 9] by providing a cross-protocol empirical baseline, and it is the first to observe Ethereum's DA market across the Fusaka (PeerDAS) and BPO transitions.

3. Protocol Architecture Comparison

This section presents a qualitative comparison of the six DA protocols along three architectural dimensions: encoding and redundancy schemes, commitment and proof systems, and verification paradigms. These dimensions collectively determine each protocol's trade-off surface between bandwidth efficiency, light client capability, trust assumptions, and fault tolerance.

3.1. Encoding and Redundancy

All six protocols employ Reed–Solomon (RS) erasure coding to introduce redundancy into block data, but they differ substantially in dimensionality, redundancy factor, and reconstruction semantics. Table 1 summarizes these differences.

The central trade-off is between redundancy overhead (bandwidth and storage cost) and reconstruction robustness. Higher

redundancy factors require proportionally more network bandwidth for data distribution and more aggregate storage across the validator or node set, but they provide stronger guarantees that data can be recovered even when a larger fraction of coded pieces is unavailable.

Celestia exhibits the highest redundancy at ∼4× due to its two-dimensional RS structure: original data of dimension $k \times k$ is extended to $2k \times 2k$, yielding four times the original data volume in shares [4]. This overhead is the direct cost of enabling two-dimensional random sampling, which allows light clients to verify availability by sampling from both rows and columns of the extended matrix. The 2D structure also permits row- or column-level reconstruction: any k shares from a single row (or column) of length $2k$ suffice to recover that row (or column), enabling localized repair without reconstructing the entire block.

Polkadot and **NEAR** both employ ∼3× redundancy, distributing erasure-coded pieces across their respective validator sets. In Polkadot's ELVES protocol, parachain block data (Proofs of Validity, or PoVs) are RS-encoded and distributed such that each validator stores one coded chunk, with reconstruction requiring $f+1$ of n chunks where $n = 3f + k$ [5, 16]. NEAR's Nightshade sharding similarly encodes chunk data with ∼3× redundancy, requiring only $1/3$ of the parts for reconstruction [7]. In both cases, the redundancy distributes data responsibility across the full validator set rather than relying on a smaller committee or random sampling.

Ethereum and **Avail** achieve the most bandwidth-efficient encoding at ∼2× redundancy. Ethereum's EIP-4844 blob transactions encode data into 128-byte cells arranged across 128 columns. PeerDAS extends each blob's polynomial evaluation points such that any 64 of the 128 columns suffice for reconstruction [3]. Avail performs row-wise RS encoding on its application data matrix, extending n original pieces to $2n$ coded pieces per row, so that any n of the $2n$ pieces suffice for recovery [8]. The 2× factor represents the minimum redundancy that still provides meaningful erasure resilience, and is sufficient for DAS-based verification when combined with appropriate commitment schemes.

Espresso's Tiramisu uses a polynomial-based Verifiable Information Dispersal (VID) scheme rather than a traditional RS code operating on byte-level shares. Data is represented as a polynomial that is evaluated at n distinct points (one per committee member); any m evaluations suffice for interpolation, where the code rate $r = m/n$ is configurable [6]. This approach is algebraically equivalent to RS coding but is tightly integrated with the VID commitment layer, enabling a unified proof of correct dispersal that simultaneously certifies both encoding correctness and data distribution.

3.2. Commitment Schemes and Proof Systems

The choice of cryptographic commitment scheme determines what properties light clients can verify, what trust assumptions are required, and the computational cost of proof generation and verification. The six protocols employ three fundamentally different commitment families: KZG polynomial commitments, Namespaced Merkle Trees (NMTs), and standard Merkle trees. Table 2 compares their properties.

Table 2: Commitment scheme properties (KZG: Avail, Ethereum, Espresso; NMT: Celestia; Merkle: Polkadot, NEAR).

Property	KZG	NMT	Merkle
Proof size	$O(1)$, ∼48 bytes	$O(\log n)$	$O(\log n)$
Trusted setup	Yes (SRS)	No	No
Namespace filter	No	Native	No
Verification	Pairings	Hashes	Hashes
Encoding correctness	Algebraic	Fraud proofs	Not verifiable

KZG polynomial commitments [14] are used by Avail, Ethereum, and Espresso. A KZG commitment to a polynomial $p(x)$ of degree d is a single elliptic curve group element (∼48 bytes on BLS12-381), and an opening proof for any evaluation $p(z) = v$ is also a single group element. Verification requires two pairing operations, which are computationally more expensive than hash evaluations but yield $O(1)$-size proofs regardless of the data size. KZG commitments enable *algebraic verification of encoding correctness*: given commitments to the rows (or columns) of an erasure-coded data matrix, anyone can verify that the extended portions are consistent with the original data by checking polynomial relationships between commitments, without accessing the underlying data. This property is what enables Avail and Ethereum light clients to reject incorrectly encoded data without fraud proofs. The principal drawback is the requirement for a structured reference string (SRS) generated through a trusted setup ceremony. Compromise of the SRS toxic waste would allow an adversary to forge proofs. In practice, both Ethereum and Avail have conducted large-scale multi-party ceremonies (Ethereum's ceremony involved over 140,000 contributors) to mitigate this risk.

Namespaced Merkle Trees (NMTs) are Celestia's defining commitment innovation [4]. An NMT is a standard Merkle tree augmented with namespace identifiers: each leaf is tagged with a namespace, internal nodes carry the minimum and maximum namespace of their subtrees, and the tree is sorted by namespace. This structure enables *namespace inclusion and exclusion proofs*. A light client can request all shares belonging to a specific namespace (e.g., a particular rollup's data) and receive a Merkle proof that the returned set is complete (no shares from that namespace have been omitted, and no extraneous shares have been included). This capability is critical for Celestia's rollup-centric design, as it allows rollup nodes to efficiently retrieve only their own data without downloading the entire block. NMT proofs are $O(\log n)$ in size. Encoding correctness, however, is *not* algebraically verifiable from the NMT alone. Instead, Celestia relies on Bad Encoding Fraud Proofs (BEFPs) to detect and propagate evidence of incorrect erasure coding [12].

Standard Merkle trees are used by Polkadot and NEAR. Polkadot commits to erasure-coded PoV chunks via a Merkle trie whose root is included in the relay chain block header [16, 31]. NEAR similarly uses Merkle roots to commit to chunk data within shard blocks. Standard Merkle proofs provide $O(\log n)$ inclusion proofs and require only hash operations for verification, making them the simplest and most widely understood commitment scheme. However, plain Merkle trees provide no

mechanism for light clients to verify encoding correctness or to perform data availability sampling: a Merkle proof confirms that a specific piece was included in the committed data set, but it reveals nothing about whether the overall erasure coding was performed correctly. In both Polkadot and NEAR, encoding correctness is enforced through the validator set (validators who reconstruct and re-encode can detect errors), but this verification is not accessible to light clients or external observers.

3.3. Verification Paradigms: DAS, VID, and Validator Recovery

The most consequential architectural distinction among DA protocols is how availability is verified. We identify three paradigms, each with different trust assumptions and light client capabilities.

3.3.1. Data Availability Sampling (DAS)

DAS, as formalized by Al-Bassam et al. [12], enables light clients to verify data availability without downloading the full block. A light client randomly requests s coded pieces from the network; if all s are returned, the client gains probabilistic confidence that the full data is available. Against a $(1-q)$-withholding adversary (one that has made fraction q of the data unavailable), a client sampling s independent pieces achieves confidence $1 - q^s$.

Celestia was the first production network to deploy DAS. Its 2D RS structure means that light clients sample from the $2k \times 2k$ extended square. Because an adversary must withhold at least ~25% of the extended square to prevent reconstruction, each sample detects withholding with probability $\geq 1/4$, so 16 random samples give a Celestia light client approximately $1 - (3/4)^{16} \approx 99\%$ confidence that the data is available, assuming that at least one honest full node is non-eclipsed and participates in the peer-to-peer network [4]. The 2D structure further enables row- or column-level reconstruction: if sampling reveals a missing row, any node holding k of the $2k$ shares in that row can reconstruct it, providing a repair mechanism that does not require full block reconstruction.

Avail implements DAS over its 1D RS-encoded rows with KZG commitments. Avail light clients sample random cells from the data matrix; each cell is accompanied by a KZG opening proof. With 10 random samples, an Avail light client achieves $1 - (1/2)^{10} \approx 99.9\%$ confidence $(1 - 1/1024)$ [8]. The use of KZG commitments provides an additional guarantee beyond sampling: the algebraic properties of polynomial commitments allow the light client to verify that each sampled cell is consistent with the committed row polynomial, effectively verifying encoding correctness without fraud proofs. Notably, Avail light clients retrieve sampled cells from a dedicated Kademlia DHT in a light-client-only P2P network before falling back to full-node RPC; cells fetched via RPC are uploaded back into the DHT, creating a self-healing replication layer that strengthens availability as light client participation grows.

Ethereum activated DAS through PeerDAS (EIP-7594) in the Fusaka upgrade on 3 December 2025. Under PeerDAS, each blob is extended from 64 to 128 evaluation points (columns), and each validator is responsible for custody of a subset of columns determined by its node ID. In the Fulu specification every node custodies at least 4 of 128 custody groups (`CUSTODY_REQUIREMENT`), nodes with attached validators custody at least 8 plus one per 32 ETH of balance (`VALIDATOR_CUSTODY_REQUIREMENT`), and every node samples 8 columns per slot (`SAMPLES_PER_SLOT`) [3]. Confidence in Ethereum's DAS is emergent at the system level rather than per-client: individual validators do not perform independent random sampling in the Celestia/Avail sense; instead, the aggregate custody across the validator set ensures that all 128 columns are held by at least one honest validator with high probability. This design reflects Ethereum's philosophy of leveraging its large validator set (~1,000,000 validators) rather than requiring each light client to independently sample. PeerDAS also introduces a "supernode" role: nodes that subscribe to all 128 data column subnets and can reconstruct the full data matrix from partial columns. The Fulu networking specification requires at least one supernode on the network for reconstruction of missing data, adding a qualitative trust assumption not present in the pre-PeerDAS regime where every consensus node downloaded all blob data [3].

The security of DAS across all three protocols rests on two critical assumptions: (i) at least one honest, non-eclipsed full node must hold the complete data and respond to sampling requests, and (ii) sufficient sampling participation must exist in the network to ensure that withholding is detected before the data propagation window closes. An adversary capable of eclipsing a light client's peer connections can cause the client to falsely conclude that data is available (or unavailable) regardless of the true state.

3.3.2. Validator-Set Recovery

Polkadot and NEAR adopt a different approach: the full validator set collectively stores and serves erasure-coded data, and availability is attested through consensus-integrated mechanisms rather than probabilistic sampling.

Polkadot's ELVES protocol tightly couples DA verification with its finality mechanism [5]. After a parachain collator produces a block candidate, the relay chain validators erasure-code the Proof of Validity (PoV) and distribute one chunk per validator. Each validator that successfully receives and stores its chunk signs an *availability bitfield* (a bitmap indicating which PoV chunks it holds). A PoV is considered available when $\geq 2/3$ of validators have signed its bitfield. Polkadot *gates finality on availability evidence*: the GRANDPA finality gadget will not finalize a relay chain block unless the availability bitfields for all included parachain candidates have reached the $2/3$ threshold [16, 31]. This design provides a deterministic (not probabilistic) availability guarantee conditioned on the $2/3$ honest-majority assumption, and ensures that unavailable data cannot be finalized.

NEAR's Nightshade sharding distributes chunk data across shard validators (chunk producers). Each chunk producer is responsible for producing and serving the chunk parts for its assigned shard. NEAR's availability enforcement relies on a

"lazy signer" mechanism with probabilistic accountability: validators are required to attest to chunk availability, but enforcement is primarily through the consensus protocol's liveness requirements rather than explicit sampling [7]. A challenge mechanism exists in the block structure for disputing invalid or unavailable chunks, but as of our observation period, this mechanism is "not used today" in practice.

The key advantage of validator-set recovery is its tight coupling with consensus: availability evidence is a prerequisite for block finality (Polkadot) or block production (NEAR), creating a direct economic incentive for validators to store and serve data. However, *light clients cannot independently verify data availability*. A Polkadot or NEAR light client must trust that the $\geq 2/3$ validator supermajority has honestly attested to availability. There is no mechanism analogous to DAS that would allow a resource-constrained client to independently check. Trust rests entirely on the validator set's honesty and the economic security provided by staking.

3.3.3. Verifiable Information Dispersal (VID)

Espresso's Tiramisu architecture introduces a third paradigm that explicitly rejects DAS in favor of algebraic dispersal proofs [6]. Tiramisu consists of three layers:

- **Savoiardi (VID layer):** Data is encoded as a polynomial and evaluated at n points (one per committee member). A vector commitment (using KZG) binds the evaluations to the polynomial commitment, producing a compact proof that each committee member received a correct evaluation. This provides *bribery resistance*: even if an adversary bribes a threshold of committee members, the algebraic binding prevents them from claiming to hold data they do not possess.
- **Mascarpone (certification layer):** Committee members who receive and verify their evaluations produce threshold signatures attesting to availability. A DA certificate is formed when $\geq 2/3$ of the committee has signed, providing a compact ($O(1)$-size) availability attestation.
- **Cocoa (CDN layer):** A secondary retrieval path through an untrusted CDN that provides availability even if committee members go offline, with cryptographic proofs linking the fallback data to the original VID commitments.

Espresso's design provides *deterministic* confidence in availability: there is no probabilistic sampling, and finalization is conditional on the DA certificate. The Savoiardi VID layer's bribery-resistance guarantees go beyond what DAS provides. While DAS detects withholding probabilistically, Savoiardi's algebraic binding makes it infeasible for a committee member to falsely attest to holding data.

The principal limitation is that data retrieval requires cooperation from the committee (or the Cocoa CDN). There is no mechanism for an independent light client to sample data and verify availability without relying on the committee's attestations. Espresso explicitly argues that DAS introduces "unnecessary redundancies" given the algebraic guarantees provided by VID [6], but this design choice means that trust in data retrievability ultimately rests on the committee and CDN infrastructure.

3.3.4. Cross-Cutting Finding: No Slashing for Data Withholding

Across all six protocols, *none implements slashing for pure data withholding*. While each protocol provides mechanisms to detect or discourage withholding, the enforcement is uniformly indirect:

- **Polkadot:** Validators are slashed for casting invalid *validity* votes (attesting that an invalid parachain block is valid), but there is no slashing for failing to store or serve erasure-coded chunks. A validator that withholds its chunk simply does not sign the availability bitfield, potentially delaying but not corrupting finality [5].
- **Celestia:** Bad Encoding Fraud Proofs (BEFPs) allow any full node to prove that the block producer performed incorrect erasure coding [12]. However, BEFPs address encoding correctness, not data withholding. A block producer that correctly encodes data but subsequently withholds it faces no direct slashing penalty.
- **Ethereum:** PeerDAS enforces availability through peer scoring and disconnection: validators that fail to serve their custody columns receive negative gossip scores and may be disconnected from the p2p network. There is no protocol-level slashing for custody failures [3].
- **Espresso:** In the current Mainnet 0 deployment, enforcement is reputational only. No slashing mechanism is implemented for committee members that fail to serve their VID evaluations [6].
- **NEAR:** A challenge mechanism exists in the block structure that theoretically allows validators to dispute chunk unavailability, but this mechanism is "not used today" in the production network. Enforcement relies on the consensus protocol's liveness requirements [7].
- **Avail:** Slashing is implemented for equivocation (double-signing) but not for data withholding. Validators that fail to make sampled cells available face no direct economic penalty [8].

This universal absence of withholding-specific slashing reflects a fundamental difficulty: proving that a node *possesses* data but *refuses* to serve it is harder than proving that a node served *incorrect* data. Data withholding is an omission fault rather than a commission fault, and attributing omission to malice (rather than network partition, latency, or node failure) remains an open problem in distributed systems.

Table 3 consolidates the architectural comparison across all dimensions discussed in this section. Several cross-cutting observations emerge. First, DAS-capable protocols (Celestia, Avail, and the emerging Ethereum PeerDAS) represent a distinct design philosophy from validator-recovery protocols (Polkadot, NEAR) and VID-based designs (Espresso). The former enable trust-minimized light clients at the cost of network-level assumptions, while the latter two provide deterministic

Table 3: Feature and parameter comparison of the six DA protocols during the observation window.

Feature	Polkadot	Ethereum	Celestia	Espresso	NEAR	Avail
Encoding	1D RS (validators)	1D RS (PeerDAS)	2D RS ($k \to 2k$)	Poly. VID	1D RS (chunks)	1D RS (rows)
Commitment	Merkle trie	KZG	NMT	KZG + vector	Merkle	KZG
Redundancy	~3×	~2×	~4×	Configurable	~3×	~2×
DAS support	No	Yes (PeerDAS)	Yes (2D)	No (VID)	No	Yes (1D)
Light client DAS	No	Custody	Yes (16 samples)	No	No	Yes (10 samples)
Block time	6 s (relay)	12 s	6 s	~2 s	~0.6 s	20 s
Max DA capacity	10 MiB×100	2.63 MiB (BPO2)	8→32 MiB*	~1 MiB	4 MiB×9	4 MiB
Protocol max MiB/s	~167	~0.22	1.33→5.31*	~0.37	~59	~0.21
Finality	12–60 s	~13 min	~6 s	~1 s	~2 s	20–60 s
DA retention	25 h	~18 d	7 d (sampling)	Node-set	~1.5 d	Node-set†
Fee model	Weight-based	Blob market	Gas-based	Per-byte	Gas-based	Weight-based
Withholding	Finality gate	Peer scoring	BEFPs (encoding)	Reputational	Unused challenge	None
Byzantine threshold	$f < n/3$	$f < n/3$	$f < n/3$	$f < n/3$	$f < n/3$/shard	$f < n/3$

*Celestia's maximum square rose from 128 to 256 (8 to 32 MiB) on 3 Dec 2025. †Avail prunes state after 256 blocks, but block data is kept by default (Substrate `archive-canonical` block pruning).

guarantees conditioned on committee or validator honesty. Second, the choice of commitment scheme is tightly coupled to the verification paradigm. KZG enables algebraic verification needed for DAS and VID, while Merkle-based schemes are sufficient for validator-recovery models where the validator set itself performs encoding verification. Third, all protocols share the same $f < n/3$ Byzantine threshold (the classical BFT bound) but differ substantially in how this threshold manifests in practice: Polkadot gates finality, Ethereum distributes custody, Celestia relies on sampling participation, Espresso requires committee certification, NEAR enforces per-shard honesty, and Avail combines sampling with algebraic commitments. These architectural choices have direct empirical consequences for throughput, cost, and operational characteristics, which we measure in Sections 4 and 5.

4. Methodology

This section details our data collection framework, metric definitions, normalization approach, and reproducibility provisions.

4.1. Data Collection Framework

We collect per-block production data from six DA protocols over a 90-day observation window (November 15, 2025 through February 14, 2026). For each protocol, we develop custom collection scripts that interface with mainnet RPC endpoints or explorer APIs, extracting block-level metrics at the granularity needed for throughput and cost analysis. All collectors are resumable: interrupted runs checkpoint progress in per-day CSV files and resume from the last collected block.

Table 4 summarizes the data sources, block counts, and collection methods for each protocol.

For each protocol, the collector captures the following:

Polkadot. We scan relay chain blocks via the Polkadot.js RPC API, extracting three event types per block: `paraInclusion.CandidateBacked`, `paraInclusion.CandidateIncluded`, and `paraInclusion.CandidateTimedOut`. We also parse the `paraInherent.enter` extrinsic to extract signed availability bitfield counts and per-core availability fractions. Throughput is computed as an upper bound: included × maxPovBytes/cadence, since the relay chain does not expose actual PoV sizes (only commitments). Cost data is collected separately from the Coretime chain, scanning `broker.Purchased`, `broker.Renewed`, and `broker.SaleInitialized` events, as well as relay chain `onDemand.OnDemandOrderPlaced` events.

Ethereum. We collect per-block header fields (`blob_gas_used`, `excess_blob_gas`) from execution-layer RPC endpoints across the full 90-day window, which spans four regimes: Pectra (target 6, max 9), Fusaka (same blob parameters, plus PeerDAS and the EIP-7918 fee floor, from block 23,935,694 on December 3, 2025), BPO1 (target 10, max 15, from block 23,975,778 on December 9), and BPO2 (target 14, max 21, from block 24,179,383 on January 7, 2026). Era boundaries are the first blocks at or after each fork's activation slot. The blob base fee is computed from the header's `excess_blob_gas` using the EIP-4844 exponential pricing formula with the update fraction of the active era. ETH/USD prices are sampled hourly from CoinGecko.

Celestia. Block data is retrieved from the Celenium indexer API, which provides 22 columns per block including `blobs_size`, `blobs_count`, `square_size`, `fill_rate`, and `fee_utia`. The window spans app version 5 and version 6 (from 24 November 2025), and a maximum-square increase from 128 to 256 (8 to 32 MiB) on 3 December 2025; we compute capacity with the maximum in force on each day.

Espresso. We fetch all blocks from the Espresso query-service explorer API using binary search to locate the start height

Table 4: Data Collection Summary Across Six DA Protocols (90-Day Window)

Protocol	Source	Blocks	Key Fields	Language	Price Source
Polkadot	Relay chain RPC	~1.3M	candidates backed/included, bitfields, core index	TypeScript	CoinGecko (DOT)
Ethereum	Execution + Beacon RPC	~648K	blob_gas_used, excess_blob_gas, blob count	Python/SQL	CoinGecko (ETH)
Celestia	Celenium API	~1.3M	blobs_size, square_size, fill_rate, fee_utia	Python	CoinGecko (TIA)
Espresso	Query-service API	~2.9M	size_bytes, num_transactions, block_time_ms	Python	CoinGecko (ETH)
NEAR	NEAR Lake S3 + RPC	~7.8M	total_encoded_bytes, gas_used, gas_price	TypeScript	CoinGecko (NEAR)
Avail	Substrate RPC	~389K	submit_data_bytes, block_fee_plancks	TypeScript	CoinGecko (AVAIL)

and 20 concurrent workers fetching 100 blocks per page (~1,630 blocks/s collection rate). Each block record contains `size_bytes`, `num_transactions`, and `block_time_ms`. Since HotShot uses responsive consensus, block times are variable (no fixed cadence). The base fee has been verified as 1 wei/byte since genesis across headers from height 1K to 10.3M.

NEAR. We employ a dual-source strategy: NEAR Lake S3 for bulk historical collection (~275 blocks/s at concurrency 100) and NEAR RPC as fallback (~2.5 blocks/s). Per-block fields include `num_shards`, `total_encoded_bytes` (sum of chunk encoded lengths), `total_gas_used`, and `gas_price`.

Avail. The collector uses the Polkadot.js API (Avail is Substrate-based) to extract `submit_data_bytes`, `submit_data_count`, and `block_fee_plancks` (from `TransactionFeePaid` events) per block. Runtime version (`spec_version`) is tracked to account for parameter changes.

4.2. Metric Definitions

We define four primary metric categories, applied consistently across all protocols.

4.2.1. Throughput (MiB/s)

We distinguish three throughput measures:

- **Protocol maximum**: the theoretical ceiling given current parameters. Protocol-specific formulas:

$$T_{\max}^{\text{Polkadot}} = \frac{\text{effective_cores} \times \text{max_pov}}{\text{cadence}} \tag{1}$$

$$T_{\max}^{\text{Ethereum}} = \frac{\text{max_blobs} \times 128\ \text{KiB}}{\text{slot_time}} \tag{2}$$

$$T_{\max}^{\text{Celestia}} = \frac{\text{max_data_bytes}}{\text{avg_block_time}} \tag{3}$$

  where effective_cores = min(num_cores, ⌊validators/5⌋) and cadence is 6s (async backing) for Polkadot.

- **Observed throughput**: measured from production data as

$$T_{\text{obs}} = \frac{\sum \text{payload_bytes}}{\Delta t \times 2^{20}} \quad \text{(MiB/s)} \tag{4}$$

  aggregated at hourly and daily granularity.

- **Utilization**: $U = T_{\text{obs}}/T_{\max}$, representing the fraction of available DA capacity in use.

4.2.2. Cost ($/MiB)

Cost normalization requires care, as protocols use fundamentally different pricing models:

- **Per-byte fee markets** (Celestia, Espresso, Avail): cost is directly computable from on-chain fee data and token prices.

$$\text{cost_per_MiB} = \text{fee_per_byte} \times 2^{20} \times \text{token_USD} \tag{5}$$

- **Blob gas market** (Ethereum): cost follows EIP-4844 exponential pricing:

$$\text{cost_per_MiB} = \frac{\text{blobBaseFee}_{\text{gwei}} \times 2^{20}}{10^9} \times \text{ETH_USD} \tag{6}$$

- **Coretime pricing** (Polkadot): DA is bundled with execution capacity. We compute:

$$\text{cost_per_MiB} = \frac{\text{coretime_price_DOT} \times \text{DOT_USD}}{\text{max_MiB_per_region}} \tag{7}$$

  for bulk coretime, and fee_paid/max_pov_MiB for on-demand. This yields a *best-case* cost assuming full PoV utilization every block.

- **Gas + storage staking** (NEAR): upload cost is gas-based (burned); state storage requires locking 1 NEAR per 100 KB. Rollup DA blobs avoid state costs by using function call inputs (pruned).

We report spot cost at percentiles (p50, p90, p99) and volume-weighted average price (VWAP) over the 90-day window.

4.2.3. Annualized Cost

Since DA is universally temporary (all protocols prune data), continuous availability requires periodic reposting. We define:

$$\text{cost_per_MiB_year} = \text{cost_per_MiB} \times \frac{365}{\text{retention_days}} \tag{8}$$

This normalization applies only where the protocol defines a retention window (Polkadot 25 h, NEAR ~1.5 d, Celestia 7 d sampling window, Ethereum ~18 d); Avail and Espresso leave retention to node operators, so we report their spot cost only.

4.2.4. Block Dynamics

We measure block time distributions (p10, p50, p90), block size distributions, and empty block rates. For protocols with variable block times (Espresso's responsive HotShot consensus), we report the empirical distribution rather than a fixed value.

4.3. Normalization and Comparability

Several methodological challenges arise from comparing heterogeneous protocols:

1. **Demand-constrained measurements**: all six protocols operate well below their theoretical capacity during our observation window (<50% utilization). Observed throughput therefore reflects *demand*, not protocol capability. We report both observed and maximum throughput and emphasize that empirical results characterize current usage patterns, not protocol limits.
2. **Token price volatility**: USD-denominated costs are sensitive to token prices, which fluctuated during our window (e.g., ETH ranged $2,000–$4,000; TIA ranged $0.30–$0.85). We use contemporaneous hourly prices from CoinGecko rather than fixed spot rates, and report VWAP alongside percentile distributions.
3. **Apples-to-oranges pricing**: Polkadot prices blockspace (coretime), not bytes. Ethereum prices blob gas. Celestia and Espresso price per byte. We normalize all to $/MiB but note that Polkadot's cost includes execution capacity (not just DA), making its per-MiB figure a lower bound on pure DA cost.
4. **Upper-bound throughput for Polkadot**: the relay chain does not expose actual PoV sizes on-chain (only commitments). Our throughput measurement uses included_candidates × max_pov_size, which is an upper bound. Actual throughput is lower since real parachains typically use 1–3 MiB per candidate.
5. **Protocol upgrades within window**: Ethereum underwent three upgrades (Fusaka, BPO1, BPO2) and Celestia transitioned eras during our observation period. We segment analysis by protocol era where applicable and treat these transitions as natural experiments (Section 5.4).
6. **Throughput measurement conventions**: Two definitional choices affect comparability with other sources (e.g., L2Beat). First, for Ethereum we report protocol maximum throughput using the *absolute maximum* blob count per block (21 under BPO2), whereas some trackers use the EIP-4844 *target* blob count (14 under BPO2), below which fees trend toward zero. Both are valid: the target represents the fee-market equilibrium, while the absolute maximum represents the hard protocol ceiling. Second, for Celestia we measure *user data throughput* (the original data square, ODS), not the full 2D Reed–Solomon encoded square. The extended square is 4× larger (a $k \times k$ ODS becomes $2k \times 2k$ after encoding), so sources that report total encoded bandwidth will show approximately 4× our figures. We report user data throughput because it reflects the amount of application-level data that can be posted per unit time.

Table 5: 90-day throughput and utilization summary.

Protocol	Max MiB/s	p50 MiB/s	Mean MiB/s	Mean util.
Polkadot	~167	~42.8[†]	~43.0[†]	~25.8%
Ethereum	0.22	~0.051	~0.051	~35%
Celestia	1.33→5.31	~0.018	~0.024	~1.7%
Espresso	0.37	~0.002	~0.003	~1.5%
NEAR	~59	~0.107	~0.108	~0.18%
Avail	0.21	~0.001	~0.002	<1%

[†]Upper bound: assumes max PoV size per included candidate. Ethereum and Celestia utilization is against the maximum in force on each day.

4.4. Reproducibility

All collection scripts, raw datasets, analysis notebooks, and generated figures are published in a public repository (`https://github.com/Chainscore/score-da`; archived at `https://doi.org/10.5281/zenodo.23276229`) under Apache 2.0 (code) and CC BY 4.0 (data) licenses. Docker environments encapsulate dependencies for each collector. Per-protocol `chain_config.json` files record exact protocol parameters, governance milestones, and era boundaries active during collection. Analysis is implemented in Python (pandas, matplotlib); scripts regenerate all figures and tables from the raw CSV data. Additionally, interactive dashboards are published for real-time exploration of the collected data.

5. Empirical Results

This section presents our findings from 90 days of production data, organized by metric.

5.1. Observation Window and Protocol Eras

Our window (November 15, 2025 to February 14, 2026) contains several parameter changes that we treat as natural experiments:

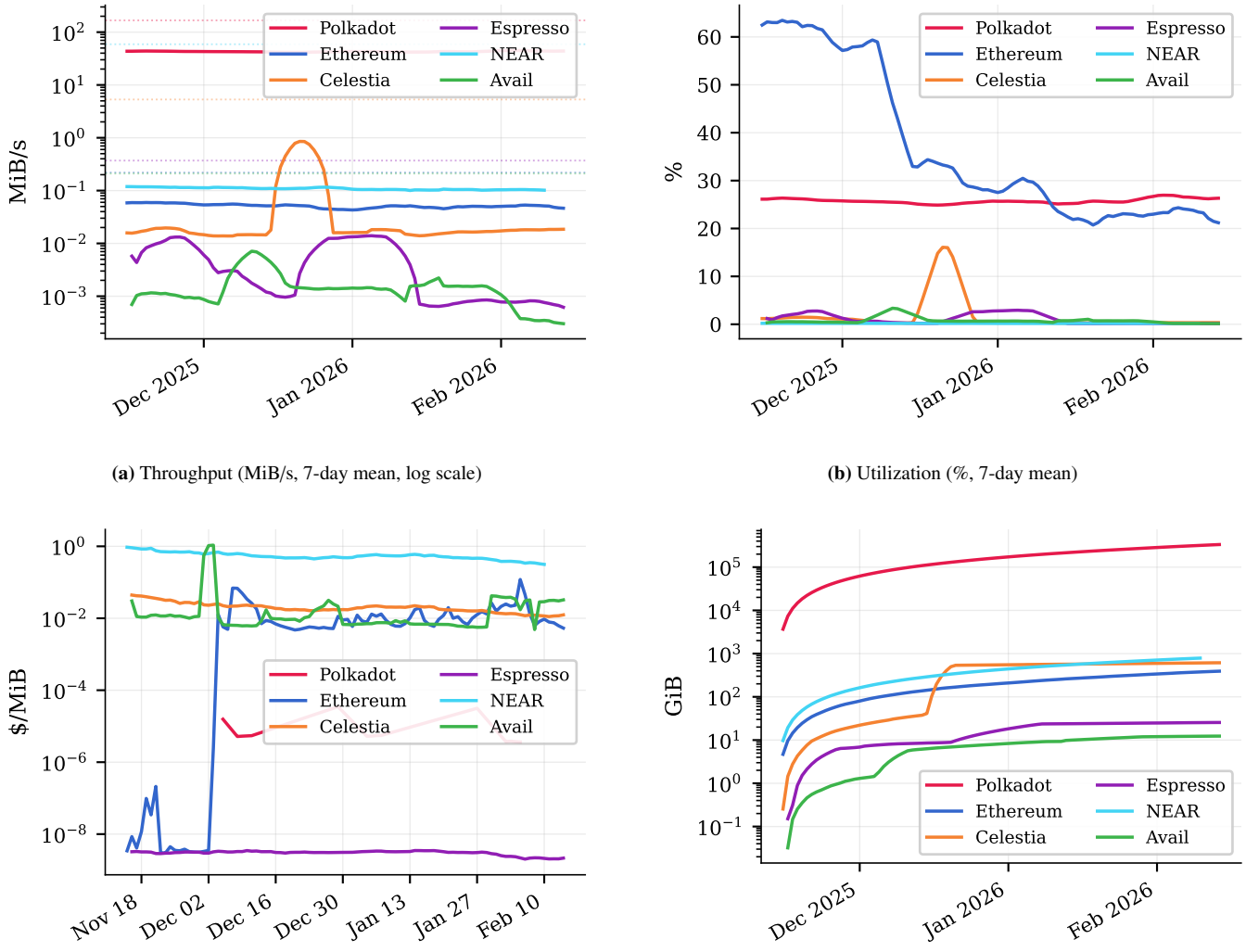
- **Ethereum**: four regimes: Pectra (target 6/max 9 blobs), Fusaka (same blob parameters, plus PeerDAS and the EIP-7918 fee floor, from Dec 3), BPO1 (target 10/max 15, from Dec 9), and BPO2 (target 14/max 21, from Jan 7).
- **Celestia**: app version 6 from Nov 24, and a maximum-square increase from 128 to 256 (8 to 32 MiB per block) on Dec 3.
- **Polkadot, Espresso, NEAR, Avail**: no major parameter changes.

Table 3 lists the active parameters.

5.2. Throughput and Utilization

Fig. 1(a) overlays 7-day rolling throughput for all six protocols on a logarithmic axis, with dashed lines at each protocol maximum; Fig. 1(b) shows utilization on a shared linear axis. Table 5 gives aggregate statistics.

Key observations:



(a) Throughput (MiB/s, 7-day mean, log scale)

(b) Utilization (%, 7-day mean)

(c) Median cost ($/MiB, log scale)

(d) Cumulative volume (GiB, log scale)

Figure 1: Cross-protocol comparison over the 90-day window. All protocols operate well below their capacity ceilings; Celestia's December 16–21 surge peaks at ~22% of its (quadrupled) capacity. Polkadot throughput and volume are upper bounds (Section 4.3).

1. **Universal under-utilization.** No protocol's mean utilization exceeds 40%. Polkadot is stable at ~25–27% (~26 of 100 cores engaged per day). Ethereum has the highest sustained utilization: ~60% under Pectra and Fusaka, falling to ~31% after BPO1 and ~23% after BPO2 as capacity grew faster than demand. Celestia runs at ≤1% in steady state, NEAR at ~0.18% despite ~9 GiB/day of payload, and Espresso and Avail at a few percent at most.
2. **Demand, not supply.** Because all protocols are demand-constrained, observed throughput reflects ecosystem adoption rather than engineering limits; raw MiB/s comparisons measure market share, not capability.
3. **NEAR's volume paradox.** NEAR has the lowest utilization but the second-highest volume (~833 GiB over the window, after Polkadot's upper-bound ~340 TiB; Ethereum ~392 GiB; Celestia ~200 GiB), because its ~59 MiB/s capacity across 9 shards far exceeds demand.

Table 6: 90-day cost summary ($/MiB).

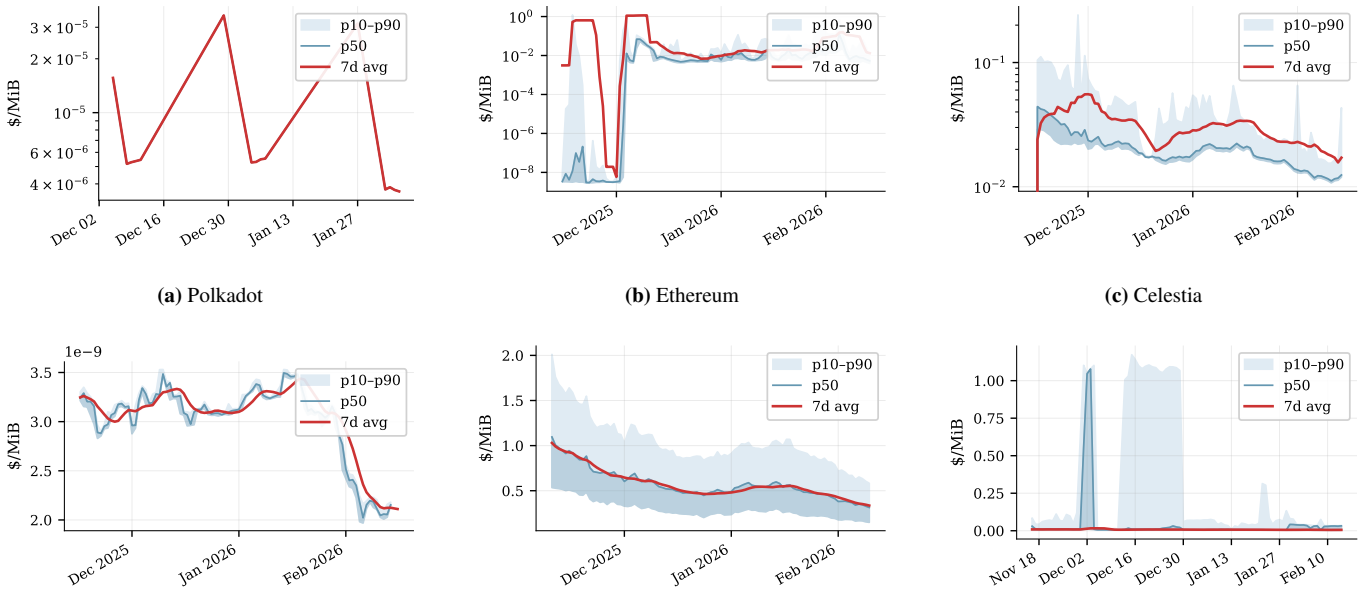
Protocol	p50	p90	p99	VWAP	Pricing
Polkadot	$5.3\times10^{-6*}$	2.7×10^{-5}	3.5×10^{-5}	—	Coretime
Ethereum	7.3×10^{-3}	0.023	0.074	0.18	Blob gas
Celestia	0.020	0.030	0.042	0.025	Gas/byte
Espresso	3.1×10^{-9}	3.4×10^{-9}	3.5×10^{-9}	3.1×10^{-9}	Fixed fee
NEAR	0.53	0.71	0.92	0.54	Gas
Avail	0.0096	0.032	1.05	0.0074	Fee+modifier

*Sparse: 14 of 94 days had bulk coretime purchases; amortized over max PoV capacity. Quantiles are over the distribution of daily median costs in the window for every protocol; VWAP is total DA spend divided by total DA volume.

5.3. Cost Analysis

Fig. 2 shows per-protocol daily cost quantile bands (p10/p50/p90) and Table 6 summarizes the distributions.

Key findings:



(a) Polkadot **(b)** Ethereum **(c)** Celestia

(d) Espresso **(e)** NEAR **(f)** Avail

Figure 2: Daily cost quantile bands ($/MiB) per protocol. (a) Polkadot bulk coretime, on days with purchases only. (b) Ethereum: fees sit at the 1-wei floor until Fusaka (Dec 3), then settle near $0.005–0.02/MiB. (c) Celestia declines from ~$0.05 to ~$0.02/MiB, partly with the TIA price. (d) Espresso's fixed 1 wei/byte fee tracks ETH/USD. (e) NEAR has the highest median among active fee markets. (f) Avail has the widest spread, with p50 spikes above $1/MiB.

1. **More than eight orders of magnitude.** Median DA cost ranges from ~$3×10^{-9}/MiB (Espresso) to ~$0.53/MiB (NEAR). The range reflects different economic designs, not only protocol efficiency.
2. **Espresso's cost is not an economic signal.** The fee has been fixed at 1 wei/byte since genesis, so its curve mirrors ETH/USD (falling from ~$3.5×10^{-9} to ~$2×10^{-9} as ETH fell). This is a permissioned Mainnet 0 artifact, not a fee market.
3. **Ethereum's floor exit is driven by EIP-7918, not by BPO demand.** Before Fusaka the blob base fee sat at the 1-wei floor for ~28% of blocks and the median cost was ~$4×10^{-9}/MiB. From the first post-Fusaka day the floor share drops to zero and median cost jumps six orders of magnitude to ~$0.009/MiB, before BPO1 adds any capacity (Section 5.4).
4. **Avail has the widest spread.** Its median ($0.0096/MiB) is comparable to Celestia's, but its p99 reaches ~$1.05 during brief activity surges around December 4 and late December, suggesting the fee modifier amplifies volatility even at low throughput.
5. **NEAR has the highest meaningful cost**, declining from ~$0.75 to ~$0.30/MiB over the window, reflecting gas pricing applied to encoded receipt data.
6. **Polkadot's cost is sparse but negligible**, because coretime bundles execution and DA; the DA component is effectively a by-product of core acquisition.

Table 7: Annualized DA cost, accounting for retention.

Protocol	Retention	Reposts/yr	p50 $ /MiB/yr
NEAR	~1.5 d	~243	~129
Celestia	7 d	~52	~1.0
Ethereum	~18 d	~20	~0.15
Polkadot	25 h	~350	~0.002
Avail	node-set	—	(spot 0.0096)
Espresso	node-set	—	(spot 3×10^{-9})

5.3.1. Retention-Adjusted Cost

Table 7 and Fig. 3(b) normalize cost by retention (Section 4.2).

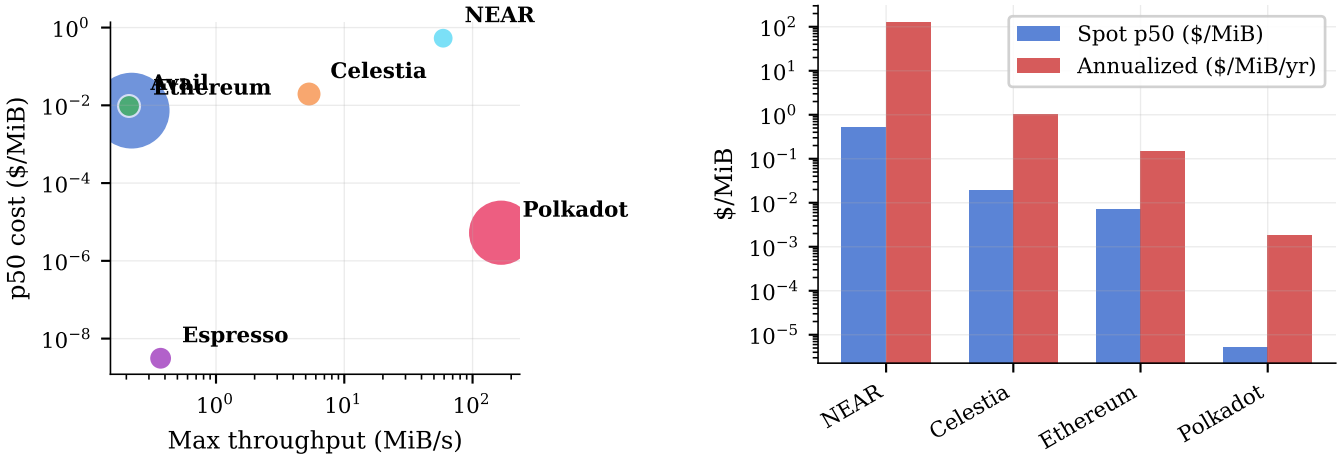
Short retention multiplies cost: NEAR's 5-epoch garbage-collection window (~1.5 days) requires ~243 reposts per year and gives by far the highest annualized cost (~$129/MiB/yr), while Celestia's 7-day sampling window lifts its $0.02/MiB spot cost to ~$1/MiB/yr. Ethereum's 18-day window keeps its annualized cost near $0.15/MiB/yr, and Polkadot's cost stays negligible even after ~350 reposts. Avail and Espresso define no protocol-level expiry, so their cost cannot be annualized on the same basis.

5.4. Natural Experiments: Protocol Upgrades

5.4.1. Ethereum: Pectra → Fusaka → BPO1 → BPO2

Fig. 4 separates the effect of Ethereum's three upgrades.

- **Fusaka (fee floor).** Under Pectra the blob fee was at the 1-wei minimum for 5–68% of blocks per day. Fusaka's EIP-



(a) Maximum throughput vs. median cost (bubble size: mean utilization)

(b) Spot vs. annualized cost

Figure 3: DA protocol landscape. Protocols cluster by design philosophy rather than along a single capacity–cost frontier, and retention reorders the cost ranking.

7918 ties a reserve price to the execution base fee; from December 4 no block is at the floor (Fig. 4(c)) and median cost rises to ~$0.009/MiB, while utilization is unchanged (~58%).

- **BPO1 and BPO2 (capacity).** Each capacity step lowered utilization (from ~58% to ~31% to ~23%) and target utilization (from ~87% to ~47% to ~34%; Fig. 4(b)). Median cost stayed in a narrow $0.007–0.010/MiB band across both BPO eras (Fig. 4(a)), consistent with fees held near the EIP-7918 reserve price rather than set by congestion.
- **Demand.** Daily blob volume stayed at ~4.1–4.7 GiB/day across all four eras, so the added capacity has not yet been absorbed.

5.4.2. *Celestia: December Demand Surge*

Celestia raised its maximum square from 128 to 256 on December 3, quadrupling capacity to ~5.3 MiB/s. From December 16 to 21, daily throughput rose eighty-fold, from a baseline of ~0.015 MiB/s to ~1.2 MiB/s. That is ~22% of the new ceiling, but it would have saturated the pre-December capacity of 1.33 MiB/s. Cost did *not* rise: the daily median fell slightly ($0.020 to $0.017/MiB) and the p90 collapsed onto the median, indicating that the surge consisted of large blobs posted at the minimum gas price rather than competing demand. Afterwards throughput and the p90 returned to their pre-event levels. The episode shows capacity being added ahead of demand: the largest demand event in our window did not exercise the fee market.

5.5. *Block Dynamics and Polkadot Core Engagement*

Fig. 5(a–b) shows block time distributions for the two protocols where they are informative; fixed-cadence protocols (Ethereum 12 s, Polkadot 6 s, Avail 20 s) are trivially tight. Celestia holds 5–6 s after an elevated p90 in mid-November. Espresso's responsive HotShot consensus has p10 ~1 s and p50 ~2 s, with a p90 that falls from ~6.5 s in the first weeks to ~3 s by February.

Fig. 5(c) shows that only ~26 of 100 Polkadot cores were engaged on a typical day, leaving ~74% idle capacity: the relay chain could host roughly three times as many parachains at current PoV sizes without parameter changes.

Espresso's payload pattern (two burst periods in November and late December–early January, then collapse to <50 MiB/day) suggests non-organic traffic typical of an early permissioned deployment, in contrast to Celestia's sustained 30–100 MiB/hour baseline.

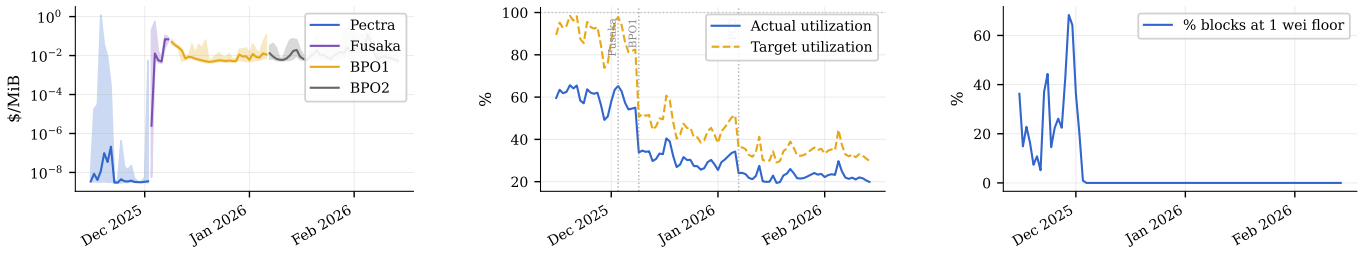
6. Security and Trust Analysis

This section analyzes the security properties and trust assumptions of each DA protocol, focusing on Byzantine fault tolerance, data withholding enforcement, retention guarantees, and decentralization characteristics.

6.1. *Byzantine Fault Tolerance*

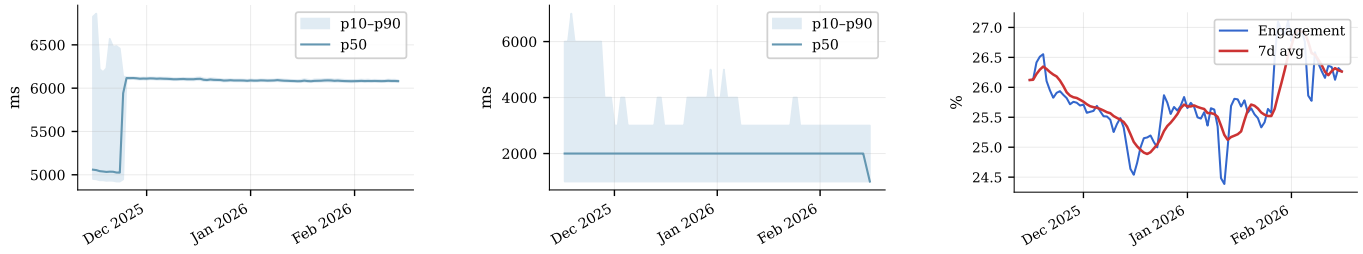
All six protocols inherit the classical BFT bound of $f < n/3$, where f is the number of Byzantine-faulty nodes and n is the total committee or validator set size. However, the practical manifestation of this threshold varies substantially.

Polkadot. Polkadot's relay chain requires $\geq 2/3$ of validators to sign availability bitfields before a parachain block can be included and finalized. With approximately 600 active validators, the system tolerates up to ~200 Byzantine faults. The ELVES protocol further strengthens this through approval checking: randomly selected validators re-execute parachain blocks post-inclusion, and a single invalid execution triggers a dispute that can slash the responsible validators [5]. The probability of an



(a) Cost bands by era ($/MiB) **(b)** Target vs. actual utilization (%) **(c)** Blocks at the 1-wei fee floor (%)

Figure 4: Ethereum blob market across the Pectra, Fusaka, BPO1 and BPO2 eras. The fee leaves the floor at Fusaka (EIP-7918), before any capacity change; the BPO forks lower utilization without materially changing median cost.



(a) Celestia block time (ms) **(b)** Espresso block time (ms) **(c)** Polkadot core engagement (%)

Figure 5: Block time distributions (p10/p50/p90) for Celestia and Espresso, and Polkadot core engagement.

invalid block escaping detection decreases exponentially with the number of approval checkers.

Ethereum. Ethereum's Casper FFG requires a 2/3 supermajority of validators (by stake weight) to finalize blocks. With approximately 1 million active validators, the economic security backing finality exceeds $30 billion (at ETH ~$3,000). PeerDAS distributes custody across the validator set: each validator holds a subset of blob columns, and system-level availability confidence emerges from the aggregate custody distribution [3]. The large validator set makes coordinated withholding attacks economically prohibitive.

Celestia. Celestia runs CometBFT (Tendermint) consensus, which requires $\geq 2/3$ of validators (by stake weight) to commit each block. The validator set is capped at 100 in the current configuration. DAS provides an additional layer of security: even if validators finalize a block with unavailable data, light client sampling will probabilistically detect the withholding [12]. This two-layer model (consensus BFT + sampling) provides stronger guarantees than consensus alone, at the cost of requiring sufficient light client participation.

Espresso. Espresso's HotShot consensus operates with 100 nodes across 20 operators in the Mainnet 0 configuration. The DA certificate requires $\geq 2/3$ of committee signatures, tolerating up to ~33 Byzantine nodes. However, the small operator count (20) means that the effective trust assumption is closer to $f < 7$ at the operator level [6].

NEAR. NEAR's Nightshade sharding applies the BFT threshold per shard. With 9 shards and approximately 400 validators, the per-shard committee size depends on stake distribution. A critical assumption is that each shard independently maintains $< 1/3$ Byzantine nodes; an adversary who concentrates stake in a single shard can potentially compromise that shard's DA at lower total cost than compromising the network-wide threshold [7]. In practice, however, most mainnet validators currently track all shards rather than only their assigned shard, which means the per-shard BFT assumption is more theoretical than operational: DA benefits from full-network replication even though the protocol is designed for shard-local storage.

Avail. Avail's BABE/GRANDPA consensus requires $\geq 2/3$ honest validators for both block production and finality. DAS light clients provide independent verification, analogous to Celestia's model. The VectorX bridge to Ethereum introduces an additional trust surface: bridge liveness depends on the prover infrastructure, and the approximately 2-hour bridge delay creates a window during which Avail-native finality and Ethereum-bridged attestations may diverge [8].

Table 8 summarizes the BFT characteristics across protocols.

6.2. Data Withholding Enforcement

As established in Section 3.3, no protocol implements slashing specifically for data withholding. Here we analyze the indirect enforcement mechanisms and their practical effectiveness.

Detection is sampling-based for Celestia, Avail and Ethereum (missing responses to random samples), attestation-

Table 8: Byzantine Fault Tolerance and Validator Set Characteristics

Protocol	BFT Threshold	Validator Count	Consensus	Finality	Stake
Polkadot	$f < n/3$	~600	BABE + GRANDPA	12–60 s	~1.8M DOT min
Ethereum	$f < n/3$ (stake)	~1M	Casper FFG	~12.8 min	32 ETH per validator
Celestia	$f < n/3$ (stake)	~100	CometBFT	Single-slot (~6 s)	Variable
Espresso	$f < n/3$	100 (20 operators)	HotShot	~1–2 s	None (permissioned)
NEAR	$f < n/3$ per shard	~400	Nightshade	~2 s	Seat-price auction
Avail	$f < n/3$	~100	BABE + GRANDPA	20–60 s	Variable

based for Polkadot and NEAR (a missing bitfield or chunk endorsement is visible on-chain, but only validator-level withholding is detected), and certificate-based for Espresso (a missing DA certificate blocks finalization).

Enforcement gap. The fundamental challenge across all protocols is the *attribution gap*: detecting that data is unavailable does not identify the responsible party. A validator that fails to serve a chunk may be offline, network-partitioned, or maliciously withholding, and the protocol cannot distinguish these cases. This attribution difficulty explains why no protocol has implemented withholding-specific slashing: the false positive rate (slashing honest nodes experiencing network issues) would be unacceptable.

6.3. Retention and Persistence

No protocol guarantees permanent availability (Table 3). Four define a window: Polkadot's availability store keeps finalized chunks for 25 hours, NEAR garbage-collects after 5 epochs (~1.5 days), Celestia light nodes sample within 7 days (CIP-036), and Ethereum serves blobs for 4,096 epochs (~18 days). Avail and Espresso define no network-wide expiry: Avail nodes keep block data by default and Espresso nodes choose their own pruning policy, so availability beyond the latest blocks rests on operators rather than the protocol. This variation has critical implications for rollup security:

- **Fraud proof windows**: Optimistic rollups require a challenge period (typically 7 days) during which fraud proofs can be submitted; the DA layer must retain data at least as long as the challenge period. Polkadot (25 h) and NEAR (~1.5 d) retain data for less than the standard 7-day fraud-proof window, and Celestia's 7-day sampling window leaves no margin, so rollups must archive data externally or shorten challenge periods; on Avail and Espresso, retention depends on node configuration.
- **Bridging latency**: Cross-chain bridges that verify DA attestations must complete verification within the retention window. Avail's VectorX bridge to Ethereum has an approximately 2-hour attestation delay, during which Avail-native finality and Ethereum-side attestations diverge.
- **Archival burden**: After DA expiry, data persistence shifts to voluntary archival by full nodes, third-party indexers, or rollup operators themselves. No protocol guarantees post-retention data retrievability.

6.4. Trust Assumptions Summary

Table 9 maps each protocol's key trust assumptions.

The trust models fall into three categories. *DAS-enhanced protocols* (Celestia, Avail, Ethereum) provide the strongest light client guarantees but require active network participation in sampling. *Validator-recovery protocols* (Polkadot, NEAR) provide deterministic guarantees conditioned on validator honesty but offer no independent light client verification. *VID-based protocols* (Espresso) provide algebraic guarantees against bribery but depend on committee and fallback infrastructure availability.

7. Discussion

This section synthesizes findings across the empirical and architectural analyses, discusses implications for protocol designers and rollup developers, and acknowledges limitations.

7.1. DAS vs. VID vs. Validator Recovery: Empirical Trade-offs

Our architectural taxonomy (Section 3) identified three verification paradigms. The empirical data reveals how these paradigms manifest in practice:

DAS protocols (Celestia, Avail, Ethereum). These protocols offer the strongest trust-minimization for light clients; sampling enables resource-constrained nodes to independently verify DA. However, DAS protocols face the *participation problem*: sampling confidence degrades if insufficient nodes participate. Celestia's December surge, an 80-fold throughput increase sustained for six days, was absorbed without a fee response because capacity had been quadrupled two weeks earlier.

Validator-recovery protocols (Polkadot, NEAR). These protocols provide the tightest coupling between DA and consensus finality. Polkadot's finality-gating mechanism (GRANDPA will not finalize blocks with insufficient availability bitfields) is the most direct enforcement mechanism we observe: unavailable data literally cannot be finalized. The trade-off is the absence of independent light client verification; trust rests entirely on the validator set. For Polkadot's use case (parachain validation), this trade-off is acceptable since the relay chain's validators are the primary consumers of DA; for general-purpose DA (serving external rollups), this model requires rollup operators to trust the validator set.

Table 9: Trust Assumption Comparison Across DA Protocols

Protocol	Availability Assumption	Encoding Trust	Additional Assumptions
Polkadot	$\geq 2/3$ validators honest (bitfield voting)	Validators re-encode and verify (approval checking)	Relay chain liveness
Ethereum	$\geq 2/3$ validators honest (Casper FFG) + sufficient custody distribution	KZG algebraic verification (no fraud proofs needed)	Trusted setup integrity (KZG ceremony)
Celestia	$\geq 2/3$ validators honest + sufficient DAS participation	BEFPs from ≥ 1 honest full node	≥ 1 non-eclipsed honest full node
Espresso	$\geq 2/3$ committee signs DA certificate	KZG + vector commitment algebraic verification	Committee honesty; CDN liveness
NEAR	$< 1/3$ Byzantine per shard	Chunk producers re-encode	Honest shard assignment; no stake concentration
Avail	$\geq 2/3$ validators honest + DAS participation	KZG algebraic verification	Trusted setup integrity; VectorX bridge liveness

VID protocol (Espresso). Espresso's Tiramisu provides algebraic bribery resistance (a property no other protocol offers), but in its current Mainnet 0 deployment, this theoretical advantage is undermined by the permissioned operator set (20 operators) and absence of slashing. The empirically observed near-zero cost (1 wei/byte) confirms that the fee market is a placeholder. As Espresso matures toward a permissionless deployment with active economic security, the VID paradigm's advantages may become more practically significant.

7.2. The Cost–Throughput Frontier

Fig. 3(a) plots DA cost against protocol maximum throughput, revealing no simple inverse relationship. Espresso offers both low cost and moderate throughput, but its cost is economically meaningless (Mainnet 0 artifact). Celestia and Avail provide moderate throughput at moderate cost. Polkadot provides the highest theoretical throughput (~167 MiB/s) at an extremely low per-MiB cost (~$\$5\times10^{-6}$/MiB), though this reflects coretime bundling execution and DA together. NEAR's high capacity (~59 MiB/s) is paired with the highest meaningful cost (~\$0.53/MiB), placing it in the opposite quadrant from Polkadot. Ethereum offers modest capacity at a cost comparable to Celestia and Avail after Fusaka, and benefits from the strongest network effects and economic security.

Figs. 1 and 3 make this heterogeneity explicit. Protocols do not cluster along a single efficiency frontier; instead, they occupy distinct positions reflecting different design philosophies: Polkadot optimizes for high-throughput parachain validation, NEAR for general-purpose smart contract execution, Ethereum for censorship resistance and economic security, and Celestia/Avail for purpose-built modular DA.

This observation underscores that *cost and throughput alone are insufficient metrics for DA evaluation*. Retention, trust assumptions, light client capabilities, and ecosystem integration are equally important, a finding that purely quantitative benchmarks risk overlooking.

7.3. Polkadot's Scaling Headroom

With only ~26 of 100 cores engaged (Fig. 5(c)), Polkadot has ~74% idle DA capacity: throughput could grow from the current upper bound of ~43 MiB/s toward ~167 MiB/s without parameter changes. As engagement rises, the coretime market should raise core prices; at the current ~$\$5\times10^{-6}$/MiB, cost is unlikely to limit near-term growth. Because actual PoV sizes are not exposed on-chain, a more precise capacity analysis requires PoV-level data.

7.4. Implications for Rollup Developers

Our findings suggest several considerations for rollup developers selecting a DA backend:

1. **Retention must exceed fraud proof windows**: Optimistic rollups with 7-day challenge periods cannot safely use DA layers with shorter retention (Polkadot: 25 h, NEAR: ~1.5 d; Celestia's 7-day sampling window leaves no margin) without external archival infrastructure. This is a binding constraint that supersedes cost considerations.
2. **Spot cost is misleading without annualization**: A rollup evaluating DA cost should compute annualized cost (Section 4.2), which accounts for the number of times data must be reposted to maintain continuous availability. Short-retention protocols that appear cheap at spot may be expensive when annualized.
3. **Fee floors, not congestion, set prices today**: Ethereum's post-Fusaka cost is pinned near the EIP-7918 reserve price, and Celestia's largest demand surge did not raise its median fee. Rollups should model protocol fee floors and token prices, not congestion, as the main cost drivers until utilization rises.
4. **Utilization headroom is currently abundant**: Every protocol's mean utilization is below 40%. In the near term, capacity is not a differentiator—demand can be accommodated by any protocol. As adoption grows and utilization approaches saturation, the empirical dynamics we measure (fee market responsiveness, throughput scaling) will become increasingly important.

7.5. Protocol Design Observations

Several cross-cutting observations emerge from comparing six production protocols simultaneously:

The withholding enforcement gap is universal. No protocol has solved the fundamental problem of attributing data withholding to specific actors. This suggests that slashing for DA omission faults may be a harder problem than slashing for commission faults (equivocation, invalid execution). Future protocol designs may need to pursue alternative enforcement models: insurance-based schemes, reputation systems with gradual stake reduction, or cryptographic mechanisms that make withholding economically irrational rather than attributably punishable.

No protocol guarantees permanence. No protocol offers permanent data availability: four define explicit expiry windows, and Avail and Espresso leave retention to node operators. This is a deliberate design choice reflecting the scalability–permanence trade-off: storing all historical data indefinitely does not scale, and DA's primary function is to provide a window for verification and fraud detection, not long-term archival. The DA layer is therefore not a substitute for archival storage; rollups and applications must implement separate data permanence strategies (e.g., IPFS, Filecoin, Arweave) for data that must be retrievable beyond the DA retention window.

Encoding redundancy vs. commitment sophistication. Protocols exhibit an inverse relationship between encoding redundancy and commitment scheme sophistication. High-redundancy encodings (Celestia's 4×, Polkadot/NEAR's 3×) use simpler commitment schemes (NMT, Merkle), while low-redundancy encodings (Ethereum/Avail's 2×) use algebraically richer commitments (KZG) that enable encoding correctness verification without fraud proofs. Espresso's polynomial VID represents a third point in this design space, using algebraic commitments to provide both encoding verification and bribery resistance with configurable redundancy.

7.6. Limitations and Threats to Validity

Demand-constrained measurements. All observed throughput figures reflect current demand, not protocol capacity. Our measurements characterize usage patterns during a specific 90-day window, not protocol performance limits under saturation. Conclusions about maximum throughput should rely on protocol parameter analysis, not observed throughput.

Token price volatility. USD-denominated costs are sensitive to the token prices observed during our window. ETH ranged from approximately \$2,000 to \$4,000, and TIA ranged from approximately \$0.30 to \$0.85. Different windows would yield different cost figures. We mitigate this by reporting VWAP and percentile distributions alongside snapshot values.

Polkadot throughput upper bound. Our Polkadot throughput measurement uses included_candidates × max_pov_size, which is an upper bound since actual PoV sizes are not exposed on-chain. Real throughput is lower; parachains typically use 1–3 MiB per PoV, not the full 10 MiB maximum.

Espresso Mainnet 0. Espresso's Mainnet 0 is a permissioned early deployment. Its cost, throughput, and security characteristics should not be compared on equal footing with permissionless, economically secured networks. We include it for architectural completeness and note its pre-production status throughout.

Single observation window. A 90-day window provides a snapshot, not a longitudinal view. Protocol behavior under different market conditions, governance events, or adoption phases may differ. Our open-source tooling enables future researchers to extend the measurement window.

Cross-protocol normalization. Comparing cost across protocols with different pricing models (blob gas, coretime, gas/byte, fixed fee) requires normalization choices that may obscure economic nuances. Polkadot's coretime bundles execution and DA, making per-MiB cost an imperfect metric. NEAR's gas pricing reflects general smart-contract costs, not DA-specific fees. Our retention-adjusted annualized cost metric (Table 7 and Fig. 3(b)) partially addresses this, but no single metric fully captures the economic trade-offs.

8. Conclusion

This paper presented the first comprehensive empirical benchmarking study of six production blockchain DA protocols (Polkadot ELVES, Ethereum EIP-4844/PeerDAS, Celestia, Espresso Tiramisu, NEAR, and Avail) using 90 days of mainnet data collected under a unified methodology.

8.1. Summary of Findings

Our analysis yields four principal findings.

First, all six protocols operate well below their theoretical capacity ceilings (<50% utilization), indicating that the current DA landscape is demand-constrained. Observed throughput measures ecosystem adoption, not protocol engineering limits. This observation is important for interpreting any cross-protocol throughput comparison: until protocols approach saturation, throughput comparisons reflect market dynamics rather than technical capability.

Second, median DA costs span more than eight orders of magnitude, from ~\$3×10^{-9}/MiB (Espresso, whose 1 wei/byte fee is a Mainnet 0 placeholder) to ~\$0.53/MiB (NEAR). On Ethereum, the exit from the 1-wei fee floor was caused by Fusaka's EIP-7918 reserve price rather than by demand attracted by the BPO capacity increases. Annualized cost normalization, which accounts for protocol-specific retention windows, reveals that protocols appearing cheap at spot pricing (due to short retention) may be expensive when continuous availability is required. The absence of a single "cheapest" protocol after annualization underscores the importance of multi-dimensional evaluation.

Third, no protocol implements slashing for pure data withholding. Enforcement is uniformly indirect: Polkadot gates finality on availability bitfields, Celestia employs bad encoding fraud proofs, Ethereum uses peer scoring, Espresso relies on

committee reputation, NEAR has an unused challenge mechanism, and Avail slashes only for equivocation. This universal gap reflects the fundamental difficulty of attributing data withholding (an omission fault) to specific actors.

Fourth, no protocol guarantees permanent availability. Protocol-defined retention ranges from 25 hours (Polkadot) to ∼18 days (Ethereum), and Avail and Espresso leave retention to node operators. Only Ethereum's window comfortably exceeds a 7-day optimistic-rollup challenge period.

8.2. Reproducibility

All collection tooling, raw datasets covering over 14 million blocks, and analysis scripts are released under Apache 2.0 (code) and CC BY 4.0 (data) licenses, with interactive dashboards for exploration.

8.3. Future Work

Several directions emerge from this study. *Longitudinal extension*: as adoption grows and protocols approach saturation, repeating this analysis will reveal supply-constrained dynamics not observable in the current demand-constrained regime. *Stress testing*: synthetic load testing under controlled conditions would complement our observational methodology with causal measurements of protocol limits. *Withholding enforcement*: the universal absence of withholding-specific slashing identifies an open problem; formal analysis of incentive-compatible enforcement mechanisms for omission faults remains an important research direction. *Cross-layer analysis*: measuring end-to-end rollup performance (confirmation time, cost, finality) as a function of DA backend choice would directly inform the rollup developer decision framework. *PeerDAS at scale*: PeerDAS activated during our window; measuring its effect on node bandwidth and on further BPO increases is a natural extension of this work.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used AI chat tools to assist with researching approaches to data collection. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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